

HAMZA MUHAMMAD

hamza@hamzamuhammad.com ◇ hamzamuhammad.com | linkedin.com/in/hamzamu

EXPERIENCE

Meta

June 2025 - present

Senior Software Engineer

- Designed and led a metric-onboarding framework that decouples new ad measurement dimensions from schema changes via a prefix-encoded slot-reuse scheme; cut onboarding time-to-market 78% (33d → 7d), unblocked 1.6% ad revenue lift across previously-blocked experiments, and eliminated a recurring class of schema-mismatch production incidents
- Built local_e2e, an end-to-end testing tool for ads measurement pipelines that compiles production SQL locally, seeds Presto with cached prod samples, and executes the DAG topologically; cut change-safety feedback loop from 2+ hours to under 30 min and eliminated test-infrastructure flakiness
- Reverse-engineered and ported a complex, sparsely-documented legacy exposure-logging system to the next-gen measurement platform; cut per-dimension storage 50x (~100TB) and raised version-pair coverage from 8% to 35%, launched with zero issues
- Designed and shipped a framework enabling the platform to join against delayed and enriched data sources (e.g. tables where today's data lands 3+ days later), unlocking 0.51% realized ads revenue lift; productionized with dashboards, alerts, and DQ checks, and cut a downstream pipeline cost 100x (\$50K+/yr savings)

Amazon

Feb 2024 - June 2025

Software Development Engineer II

- Designed and shipped React Native mobile app replacing a desktop tool for real-time security monitoring across 175+ Amazon fulfillment centers; serves 3K+ daily on-site staff and cut incident response time from hours to minutes
- Built and shipped to production a RAG-based resolution agent over historical support-case data, autonomously handling ~17% of ~1K monthly security tickets (access requests, alarm faults, etc.) and reducing on-call toil for routine triage

Dropbox

July 2022 - Feb 2024

Senior Software Engineer

- Contributed to retrieval architecture for Dropbox Dash combining dense/lexical search with knowledge-graph traversal, shipped as the default ranking stack at launch
- Owned ingestion and indexing pipeline for Dash supporting GA launch to thousands of users across enterprise partners (incl. McLaren Racing); sustained 10K docs/sec with p99 freshness of 7min, solving backpressure around embedding-generation bottleneck via adaptive rate-limiting and chunking
- Drove hybrid-cloud architecture connecting on-prem Kubernetes to AWS via Direct Connect, unlocking managed services (MSK, RDS) for Platform org; adopted as the reference pattern for subsequent migrations

Amazon

Oct 2019 - July 2022

Software Development Engineer

- Built continuous-training framework and multi-tenant data lake for ML inference and A/B evaluation; productionized NLP/CV models behind high-scale recommendation surfaces

SKILLS

Languages	Python, Golang, TypeScript, Java, Scala, GraphQL, SQL
Frameworks	PyTorch, Spark, Terraform, CDK, React Native, Pandas
Platforms	Kafka, Elasticsearch, Redis, Docker, Bazel, AWS

EDUCATION

The University of Texas at Austin
B.S. Computer Science, with Honors

May 2018